Bayesian Optimization Report: Function 2

GP Health Summary per Iteration

Iteration	GP Health	Status
1	0.749	Good
2	0.749	Good
3	0.748	Good
4	0.748	Good
5	0.748	Good
6	0.748	Good
7	0.748	Good
8	0.747	Good
9	0.747	Good
10	0.747	Good
11	0.746	Good
12	0.746	Good
13	0.745	Good
14	0.745	Good
15	0.744	Good
16	0.744	Good
17	0.743	Good
18	0.743	Good
19	0.742	Good
20	0.742	Good
21	0.742	Good
22	0.740	Good
23	0.740	Good
24	0.739	Good
25	0.739	Good
26	0.738	Good
27	0.737	Good
28	0.737	Good
29	0.736	Good
30	0.736	Good

GP Sigma Summary

Iteration	Mean Sigma	Min Sigma	Max Sigma
1	0.2380	0.2378	0.2380
2	0.2309	0.2299	0.2310
3	0.2246	0.2242	0.2247
4	0.2185	0.2163	0.2187
5	0.2122	0.2056	0.2132
6	0.2066	0.1978	0.2081
7	0.2016	0.1907	0.2034
8	0.1965	0.1874	0.1989
9	0.1909	0.1777	0.1948
10	0.1862	0.1718	0.1908
11	0.1846	0.1721	0.1877
12	0.1797	0.1676	0.1843
13	0.1757	0.1590	0.1809
14	0.1735	0.1572	0.1791
15	0.1683	0.1435	0.1765
16	0.1636	0.1398	0.1736
17	0.1618	0.1390	0.1709
18	0.1561	0.1310	0.1682
19	0.1498	0.1201	0.1657
20	0.1473	0.1207	0.1632
21	0.1425	0.1144	0.1609
22	0.1480	0.1292	0.1597
23	0.1452	0.1277	0.1575
24	0.1423	0.1216	0.1560
25	0.1403	0.1229	0.1540
26	0.1367	0.1174	0.1521
27	0.1335	0.1146	0.1500
28	0.1304	0.1116	0.1483
29	0.1209	0.0934	0.1465
30	0.1175	0.0930	0.1431

Best Results Per Iteration

Iteration	Best Input	Best Output	Kernel	Acquisition	GP Health	Anomalies
1	[0.63509365 0.04530401]	0.29099395343776285	Matern(ν=0.5)	UCB	0.749	None
2	[0.74826113 0.84696538]	0.2957536333844084	Matern(ν=0.5)	UCB	0.749	None
3	[0.2251558 0.95371151]	0.28522250658233833	Matern(ν=0.5)	UCB	0.748	None
4	[0.2597404 0.16215267]	0.2852113992103971	Matern(ν=0.5)	UCB	0.748	None
5	[0.50534251 0.97758644]	0.28476408140943565	Matern(ν=0.5)	UCB	0.748	None
6	[0.07514555 0.88598883]	0.28483735934855764	Matern(ν=0.5)	UCB	0.748	None
7	[0.3946126 0.1399748]	0.28443907986282985	Matern(ν=0.5)	UCB	0.748	None
8	[0.19272004 0.82813138]	0.2840717065724476	Matern(ν=0.5)	UCB	0.747	None
9	[0.28839886 0.05056745]	0.284225261350961	Matern(ν=0.5)	UCB	0.747	None
10	[0.65280448 0.66587567]	0.3665493631664476	Matern(ν=0.5)	UCB	0.747	None
11	[0.80766356 0.58195819]	0.2880405703957225	Matern(ν=0.5)	UCB	0.746	None
12	[0.61032472 0.01159321]	0.2833200681699656	Matern(ν=0.5)	UCB	0.746	None
13	[0.67808695 0.72407176]	0.4925442015699764	Matern(ν=0.5)	UCB	0.745	None
14	[0.86411417 0.84275202]	0.3457973957311157	Matern(ν=0.5)	UCB	0.745	None
15	[0.94504827 0.39425979]	0.2954725619083653	Matern(ν=1.5)	UCB	0.744	None
16	[0.10219511 0.45017733]	0.29406221761558243	Matern(ν=1.5)	UCB	0.744	None
17	[0.04798212 0.82203608]	0.29413971166299685	Matern(ν=2.5)	UCB	0.743	None
18	[0.53213611 0.52925127]	0.29350562407670067	Matern(ν=2.5)	UCB	0.743	None
19	[0.77790251 0.61295406]	0.29531329577382165	Matern(ν=2.5)	UCB	0.742	None
20	[0.48287587 0.36992832]	0.2927547410545262	Matern(ν=2.5)	UCB	0.742	None
21	[0.71364922 0.3410387]	0.4700616828386541	Matern(ν=2.5)	UCB	0.742	None

22	[0.92352274 0.67699633]	0.2939306150661813	Matern(ν=0.5)	UCB	0.740	None
23	[0.73357796 0.27443329]	0.3635173174369073	Matern(ν=0.5)	UCB	0.740	None
24	[0.37209783 0.694409]	0.29363291425000554	Matern(ν=0.5)	UCB	0.739	None
25	[0.41525689 0.43662749]	0.29522023888680066	Matern(ν=0.5)	UCB	0.739	None
26	[0.96631801 0.37199329]	0.2935574866160415	Matern(ν=0.5)	UCB	0.738	None
27	[0.82804416 0.23890489]	0.29795262693249197	Matern(ν=0.5)	UCB	0.737	None
28	[0.30588298 0.9803708]	0.291834890056845	Matern(ν=0.5)	UCB	0.737	None
29	[0.17149682 0.09645316]	0.2892199315276769	Matern(ν=1.5)	UCB	0.736	None
30	[0.2425356 0.19137025]	0.29451589544621304	Matern(ν=1.5)	UCB	0.736	None

Acquisition Values Per Iteration

Iteration 1

Method	Max Value
UCB	0.770239
EI	0.001963
PI	0.021437
THOMPSON	0.643312
KG	0.001963
ENTROPY	1.391272
PORTFOLIO	0.966959

Iteration 2

Method	Max Value
UCB	0.429162
EI	0.001710
PI	0.019525
THOMPSON	0.739323
KG	0.001710
ENTROPY	1.371258

PORTFOLIO	0.868740

Iteration 3

Method	Max Value
UCB	0.416119
EI	0.001288
PI	0.015539
THOMPSON	0.648186
KG	0.001288
ENTROPY	1.322368
PORTFOLIO	0.863156

Iteration 4

Method	Max Value
UCB	0.410649
EI	0.001077
PI	0.013572
THOMPSON	0.617503
KG	0.001077
ENTROPY	1.314569
PORTFOLIO	0.778653

Iteration 5

Method	Max Value
UCB	0.405569
EI	0.000903
PI	0.011864
THOMPSON	0.562333
KG	0.000903
ENTROPY	1.232737
PORTFOLIO	0.729028

Iteration 6

Method	Max Value
UCB	0.400800

EI	0.000759
PI	0.010377
THOMPSON	0.724767
KG	0.000759
ENTROPY	1.261134
PORTFOLIO	0.670833

Iteration 7

Method	Max Value
UCB	0.432391
EI	0.001005
PI	0.014420
THOMPSON	0.723002
KG	0.001005
ENTROPY	1.195623
PORTFOLIO	0.674904

Iteration 8

Method	Max Value
UCB	0.421102
EI	0.000762
PI	0.011424
THOMPSON	0.626629
KG	0.000762
ENTROPY	1.219003
PORTFOLIO	0.605706

Iteration 9

Method	Max Value
UCB	0.440101
EI	0.000953
PI	0.014546
THOMPSON	0.591926
KG	0.000953
ENTROPY	1.269023

PORTFOLIO	0.575427

Iteration 10

Method	Max Value
UCB	0.437179
EI	0.000838
PI	0.013214
THOMPSON	0.663887
KG	0.000838
ENTROPY	1.245174
PORTFOLIO	0.582518

Iteration 11

Method	Max Value
UCB	0.416953
EI	0.000528
PI	0.008793
THOMPSON	0.581307
KG	0.000528
ENTROPY	1.215326
PORTFOLIO	0.524298

Iteration 12

Method	Max Value
UCB	0.416348
EI	0.000457
PI	0.007909
THOMPSON	0.584508
KG	0.000457
ENTROPY	1.209831
PORTFOLIO	0.547402

Iteration 13

Method	Max Value
UCB	0.510346

EI	0.002962
PI	0.043306
THOMPSON	0.581778
KG	0.002962
ENTROPY	1.082155
PORTFOLIO	0.491640

Iteration 14

Method	Max Value
UCB	0.408383
EI	0.000381
PI	0.006746
THOMPSON	0.646978
KG	0.000381
ENTROPY	1.187183
PORTFOLIO	0.473042

Iteration 15

Method	Max Value
UCB	0.402593
EI	0.000234
PI	0.004144
THOMPSON	0.572428
KG	0.000234
ENTROPY	1.105747
PORTFOLIO	0.451671

Iteration 16

Method	Max Value
UCB	0.394017
EI	0.000200
PI	0.003654
THOMPSON	0.559465
KG	0.000200
ENTROPY	1.108939

PORTFOLIO	0.414619

Iteration 17

Method	Max Value
UCB	0.395041
EI	0.000173
PI	0.003246
THOMPSON	0.514099
KG	0.000173
ENTROPY	1.111241
PORTFOLIO	0.418377

Iteration 18

Method	Max Value
UCB	0.392516
EI	0.000149
PI	0.002865
THOMPSON	0.566022
KG	0.000149
ENTROPY	1.064291
PORTFOLIO	0.382426

Iteration 19

Method	Max Value
UCB	0.396499
EI	0.000129
PI	0.002542
THOMPSON	0.467995
KG	0.000129
ENTROPY	1.057999
PORTFOLIO	0.406906

Iteration 20

Method	Max Value
UCB	0.387686

EI	0.000102
PI	0.002089
THOMPSON	0.493227
KG	0.000102
ENTROPY	1.054776
PORTFOLIO	0.375940

Iteration 21

Method	Max Value
UCB	0.452259
EI	0.000270
PI	0.006905
THOMPSON	0.470318
KG	0.000270
ENTROPY	1.039486
PORTFOLIO	0.355241

Iteration 22

Method	Max Value
UCB	0.381361
EI	0.000083
PI	0.001755
THOMPSON	0.513802
KG	0.000083
ENTROPY	1.053404
PORTFOLIO	0.343434

Iteration 23

Method	Max Value
UCB	0.388133
EI	0.000070
PI	0.001687
THOMPSON	0.528544
KG	0.000070
ENTROPY	1.106383

PORTFOLIO	0.372947

Iteration 24

Method	Max Value
UCB	0.356200
EI	0.000064
PI	0.001423
THOMPSON	0.603265
KG	0.000064
ENTROPY	1.082186
PORTFOLIO	0.367903

Iteration 25

Method	Max Value
UCB	0.354175
EI	0.000055
PI	0.001239
THOMPSON	0.507746
KG	0.000055
ENTROPY	1.067237
PORTFOLIO	0.322836

Iteration 26

Method	Max Value
UCB	0.352207
EI	0.000048
PI	0.001104
THOMPSON	0.519116
KG	0.000048
ENTROPY	0.958984
PORTFOLIO	0.325235

Iteration 27

Method	Max Value
UCB	0.350047

EI	0.000037
PI	0.000888
THOMPSON	0.431100
KG	0.000037
ENTROPY	0.995108
PORTFOLIO	0.295343

Iteration 28

Method	Max Value
UCB	0.344896
EI	0.000032
PI	0.000773
THOMPSON	0.456196
KG	0.000032
ENTROPY	0.975847
PORTFOLIO	0.298395

Iteration 29

Method	Max Value
UCB	0.343400
EI	0.000028
PI	0.000691
THOMPSON	0.510542
KG	0.000028
ENTROPY	0.955077
PORTFOLIO	0.286257

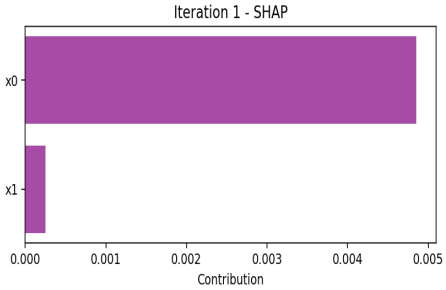
Iteration 30

Method	Max Value
UCB	0.340376
EI	0.000018
PI	0.000478
THOMPSON	0.426035
KG	0.000018
ENTROPY	0.925250

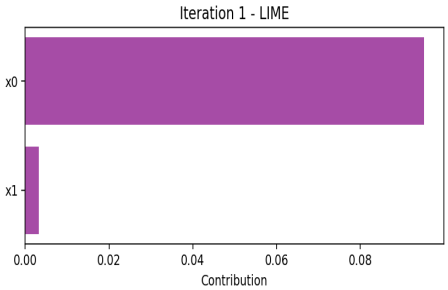
PORTFOLIO	0.292188

Feature Explanations (SHAP / LIME)

Iteration 1 - SHAP



Iteration 1 - LIME



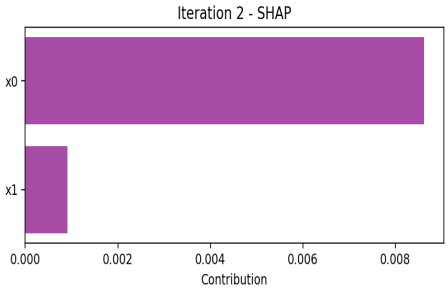
Iteration 1 - SHAP Values

Feature	SHAP Value
x0	0.00485
x1	0.00025

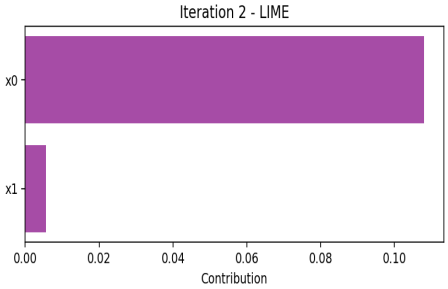
Iteration 1 - LIME Values

Feature	LIME Value
x0	0.09522
x1	0.00328

Iteration 2 - SHAP



Iteration 2 - LIME



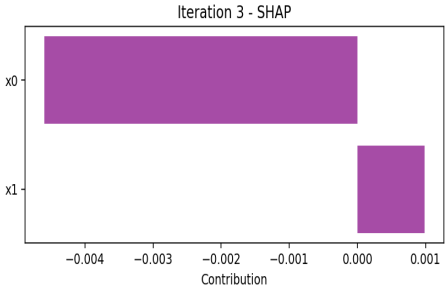
Iteration 2 - SHAP Values

Feature	SHAP Value
x0	0.00862
x1	0.00091

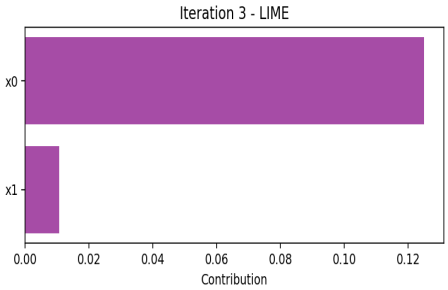
Iteration 2 - LIME Values

Feature	LIME Value
x0	0.10795
x1	0.00553

Iteration 3 - SHAP



Iteration 3 - LIME



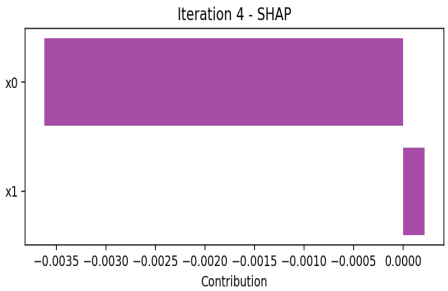
Iteration 3 - SHAP Values

Feature	SHAP Value
x0	-0.00460
x1	0.00099

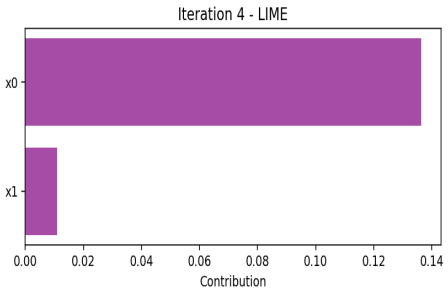
Iteration 3 - LIME Values

Feature	LIME Value
x0	0.12488
x1	0.01063

Iteration 4 - SHAP



Iteration 4 - LIME



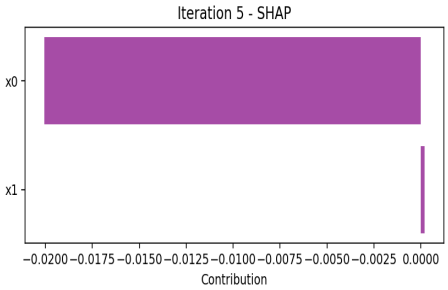
Iteration 4 - SHAP Values

Feature	SHAP Value
x0	-0.00362
x1	0.00022

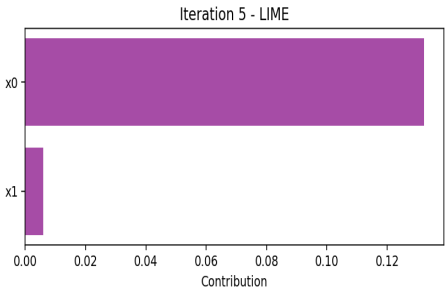
Iteration 4 - LIME Values

Feature	LIME Value
x0	0.13631
x1	0.01089

Iteration 5 - SHAP



Iteration 5 - LIME



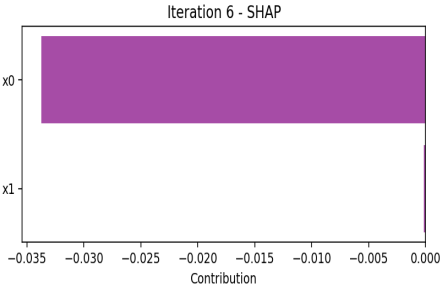
Iteration 5 - SHAP Values

Feature	SHAP Value
x0	-0.02006
x1	0.00021

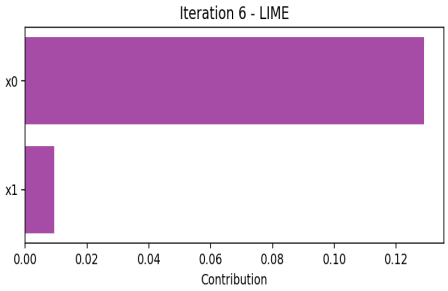
Iteration 5 - LIME Values

Feature	LIME Value
x0	0.13217
x1	0.00602

Iteration 6 - SHAP



Iteration 6 - LIME



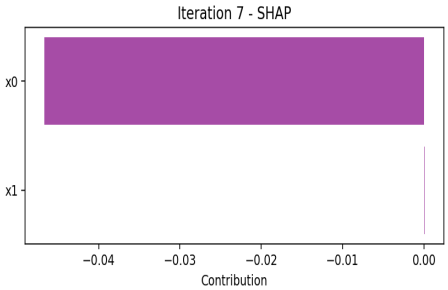
Iteration 6 - SHAP Values

Feature	SHAP Value
x0	-0.03373
x1	-0.00015

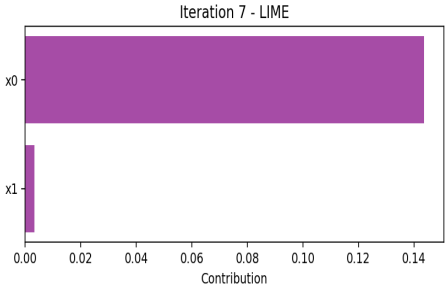
Iteration 6 - LIME Values

Feature	LIME Value
x0	0.12895
x1	0.00942

Iteration 7 - SHAP



Iteration 7 - LIME



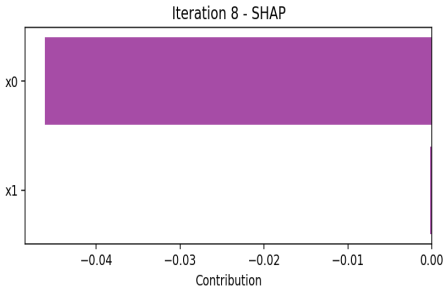
Iteration 7 - SHAP Values

Feature	SHAP Value
x0	-0.04666
x1	0.00009

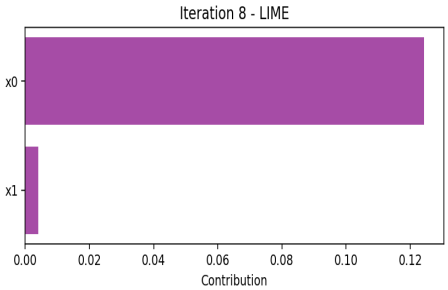
Iteration 7 - LIME Values

Feature	LIME Value
x0	0.14360
x1	0.00337

Iteration 8 - SHAP



Iteration 8 - LIME



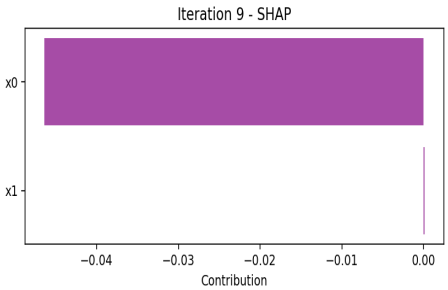
Iteration 8 - SHAP Values

Feature	SHAP Value
x0	-0.04614
x1	-0.00025

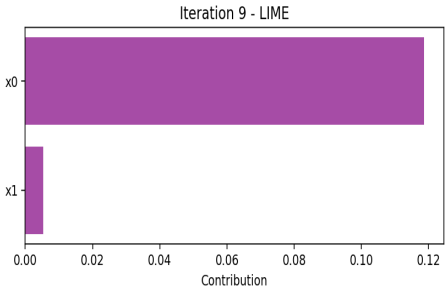
Iteration 8 - LIME Values

Feature	LIME Value
x0	0.12425
x1	0.00408

Iteration 9 - SHAP



Iteration 9 - LIME



Iteration 9 - SHAP Values

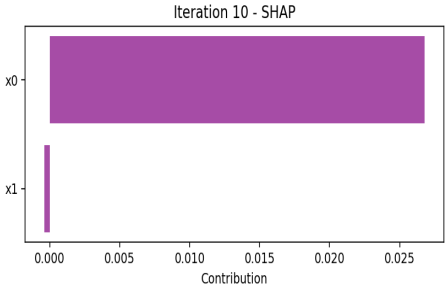
Feature	SHAP Value
x0	-0.04640
x1	0.00015

Iteration 9 - LIME Values

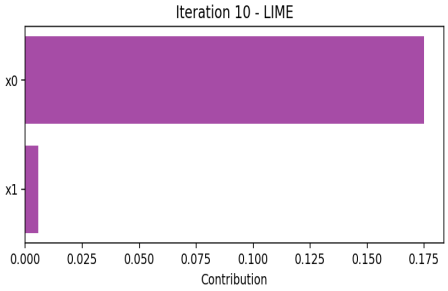
Feature	LIME Value
x0	0.11861

x1	0.00532

Iteration 10 - SHAP



Iteration 10 - LIME



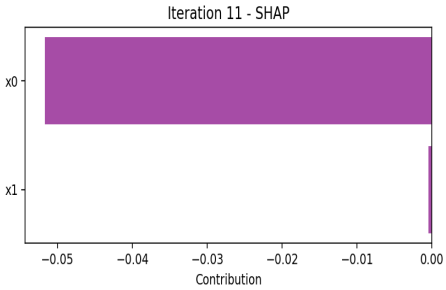
Iteration 10 - SHAP Values

Feature	SHAP Value
x0	0.02680
x1	-0.00038

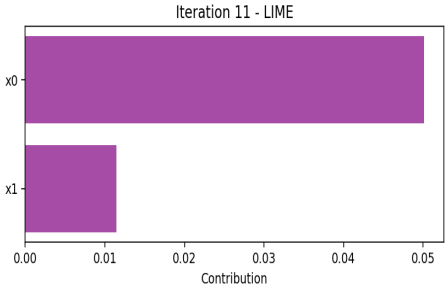
Iteration 10 - LIME Values

Feature	LIME Value
x0	0.17492
x1	0.00568

Iteration 11 - SHAP



Iteration 11 - LIME



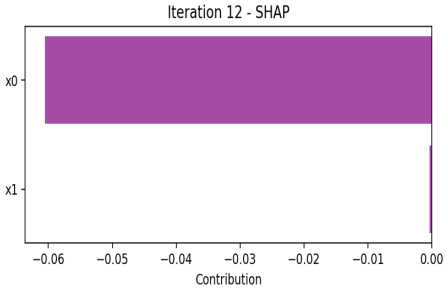
Iteration 11 - SHAP Values

Feature	SHAP Value
x0	-0.05173
x1	-0.00045

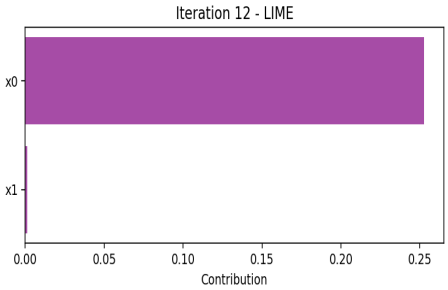
Iteration 11 - LIME Values

Feature	LIME Value
x0	0.05011
x1	0.01142

Iteration 12 - SHAP



Iteration 12 - LIME



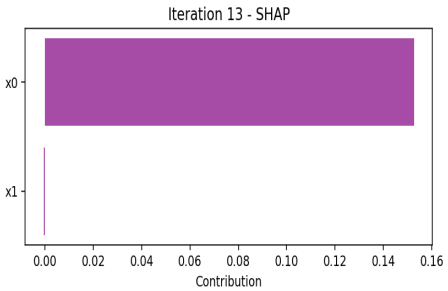
Iteration 12 - SHAP Values

Feature	SHAP Value
x0	-0.06059
x1	-0.00040

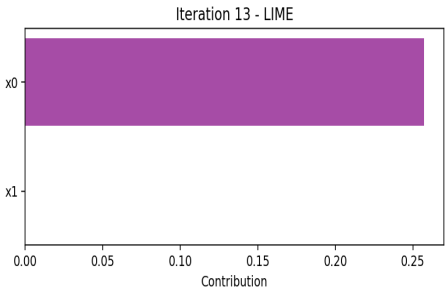
Iteration 12 - LIME Values

Feature	LIME Value
x0	0.25251
x1	0.00123

Iteration 13 - SHAP



Iteration 13 - LIME



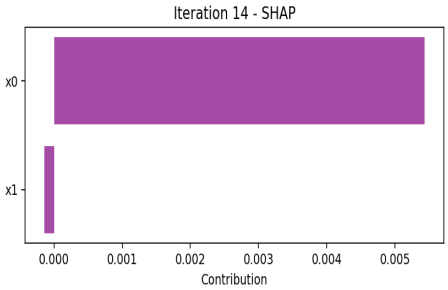
Iteration 13 - SHAP Values

Feature	SHAP Value
x0	0.15262
x1	-0.00045

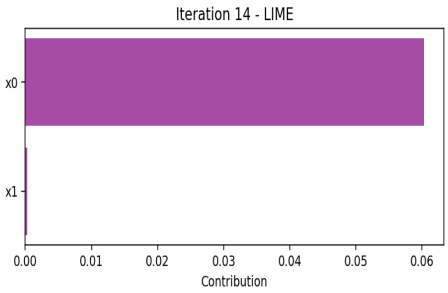
Iteration 13 - LIME Values

Feature	LIME Value
x0	0.25708
x1	0.00000

Iteration 14 - SHAP



Iteration 14 - LIME



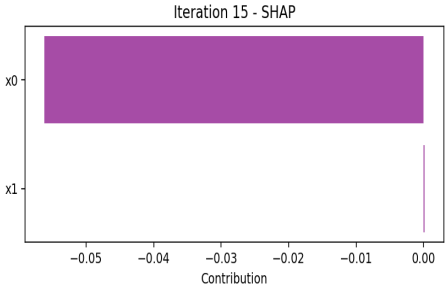
Iteration 14 - SHAP Values

Feature	SHAP Value
x0	0.00544
x1	-0.00014

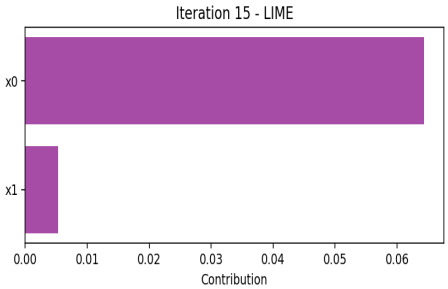
Iteration 14 - LIME Values

Feature	LIME Value
x0	0.06027
x1	0.00026

Iteration 15 - SHAP



Iteration 15 - LIME



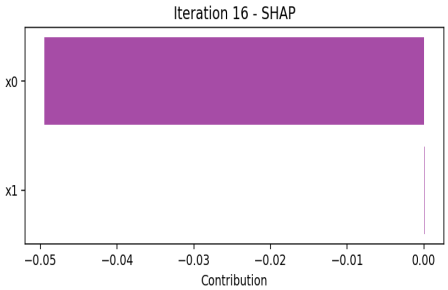
Iteration 15 - SHAP Values

Feature	SHAP Value
x0	-0.05621
x1	0.00016

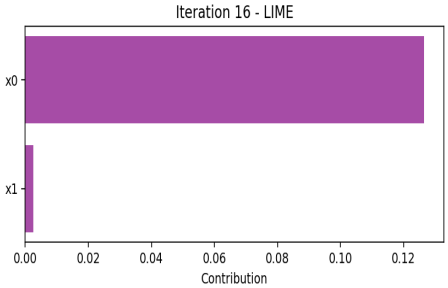
Iteration 15 - LIME Values

Feature	LIME Value
x0	0.06433
x1	0.00527

Iteration 16 - SHAP



Iteration 16 - LIME



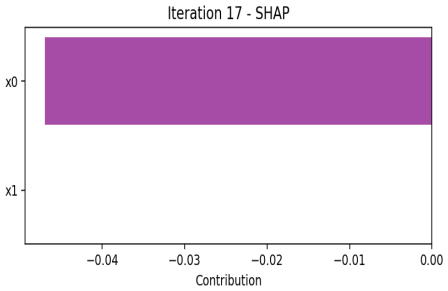
Iteration 16 - SHAP Values

Feature	SHAP Value
x0	-0.04951
x1	0.00012

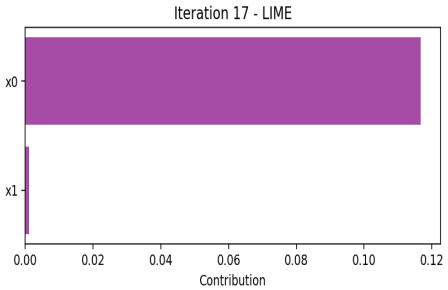
Iteration 16 - LIME Values

Feature	LIME Value
x0	0.12644
x1	0.00246

Iteration 17 - SHAP



Iteration 17 - LIME



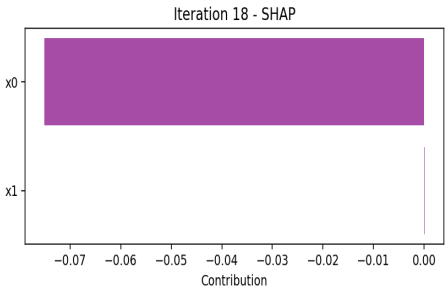
Iteration 17 - SHAP Values

Feature	SHAP Value
x0	-0.04698
x1	-0.00003

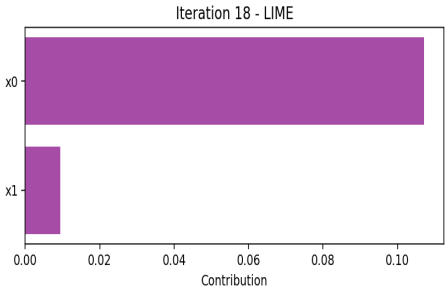
Iteration 17 - LIME Values

Feature	LIME Value
x0	0.11678
x1	0.00101

Iteration 18 - SHAP



Iteration 18 - LIME



Iteration 18 - SHAP Values

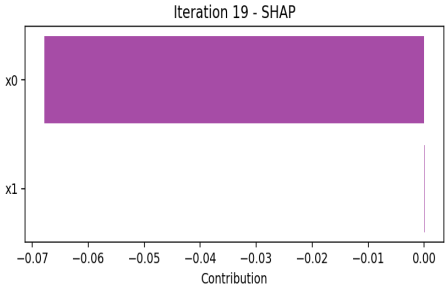
Feature	SHAP Value
x0	-0.07510
x1	0.00014

Iteration 18 - LIME Values

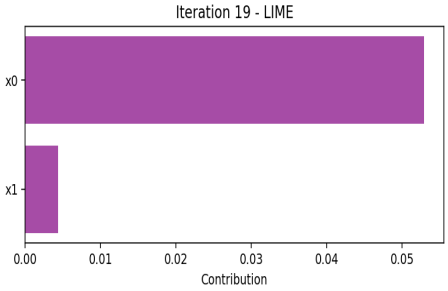
Feature	LIME Value
x0	0.10712

x1	0.00943

Iteration 19 - SHAP



Iteration 19 - LIME



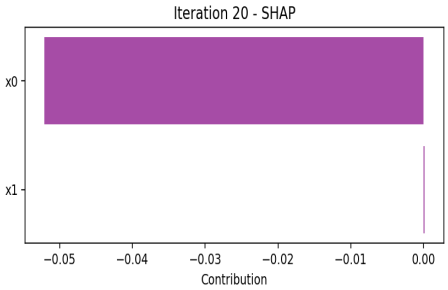
Iteration 19 - SHAP Values

Feature	SHAP Value
x0	-0.06786
x1	0.00009

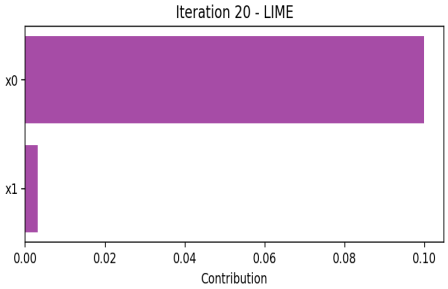
Iteration 19 - LIME Values

Feature	LIME Value
x0	0.05289
x1	0.00433

Iteration 20 - SHAP



Iteration 20 - LIME



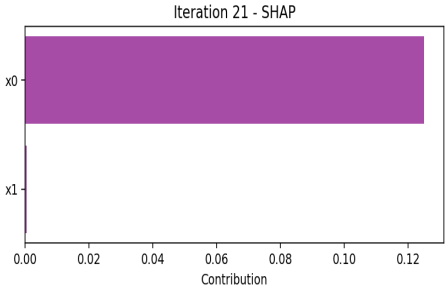
Iteration 20 - SHAP Values

Feature	SHAP Value
x0	-0.05211
x1	0.00018

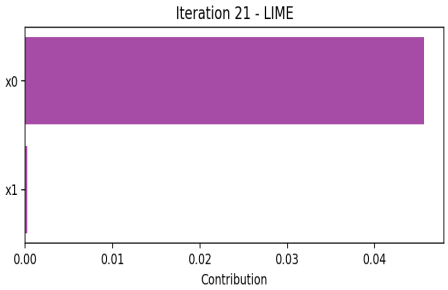
Iteration 20 - LIME Values

Feature	LIME Value
x0	0.09988
x1	0.00310

Iteration 21 - SHAP



Iteration 21 - LIME



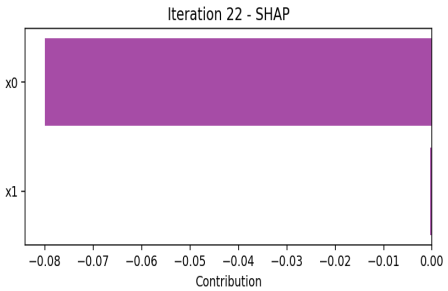
Iteration 21 - SHAP Values

Feature	SHAP Value
x0	0.12497
x1	0.00040

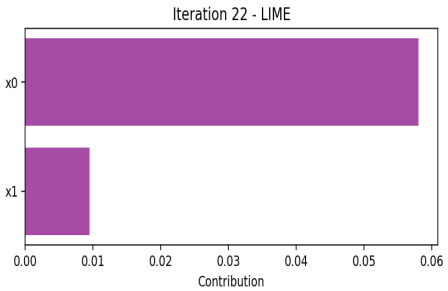
Iteration 21 - LIME Values

Feature	LIME Value
x0	0.04562
x1	0.00020

Iteration 22 - SHAP



Iteration 22 - LIME



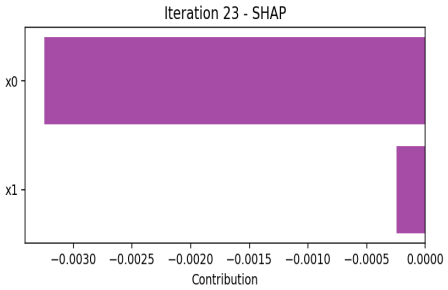
Iteration 22 - SHAP Values

Feature	SHAP Value
x0	-0.08003
x1	-0.00037

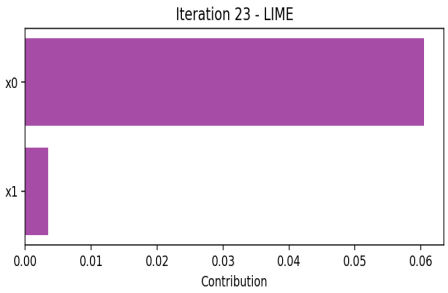
Iteration 22 - LIME Values

Feature	LIME Value
x0	0.05803
x1	0.00952

Iteration 23 - SHAP



Iteration 23 - LIME



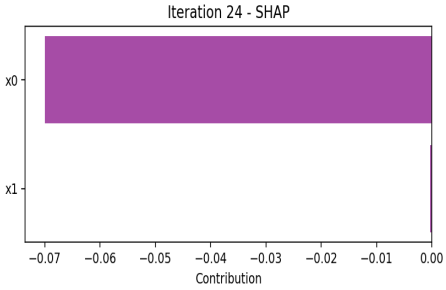
Iteration 23 - SHAP Values

Feature	SHAP Value
x0	-0.00325
x1	-0.00024

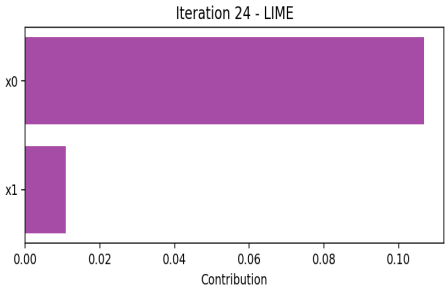
Iteration 23 - LIME Values

Feature	LIME Value
x0	0.06046
x1	0.00350

Iteration 24 - SHAP



Iteration 24 - LIME



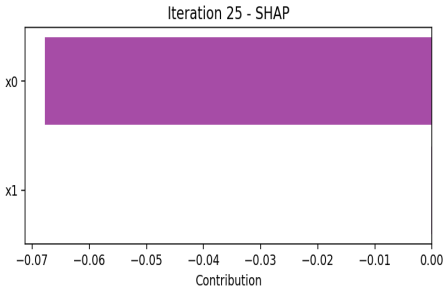
Iteration 24 - SHAP Values

Feature	SHAP Value
x0	-0.07003
x1	-0.00034

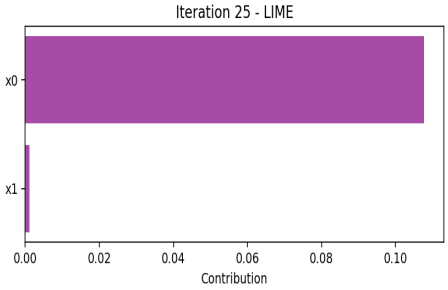
Iteration 24 - LIME Values

Feature	LIME Value
x0	0.10679
x1	0.01080

Iteration 25 - SHAP



Iteration 25 - LIME



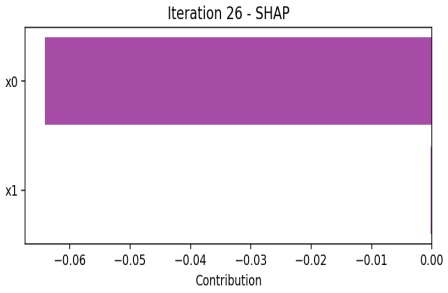
Iteration 25 - SHAP Values

Feature	SHAP Value
x0	-0.06782
x1	-0.00018

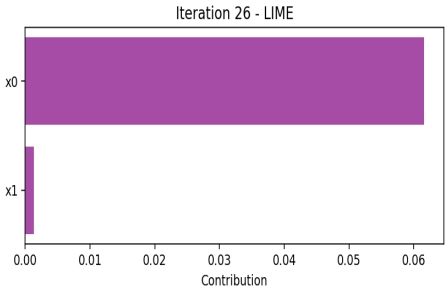
Iteration 25 - LIME Values

Feature	LIME Value
x0	0.10764
x1	0.00118

Iteration 26 - SHAP



Iteration 26 - LIME



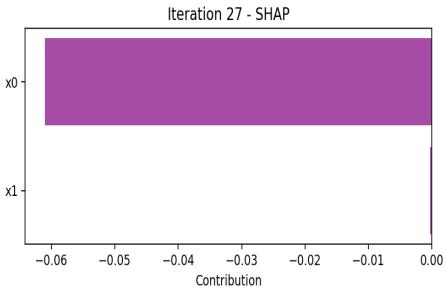
Iteration 26 - SHAP Values

Feature	SHAP Value
x0	-0.06416
x1	-0.00023

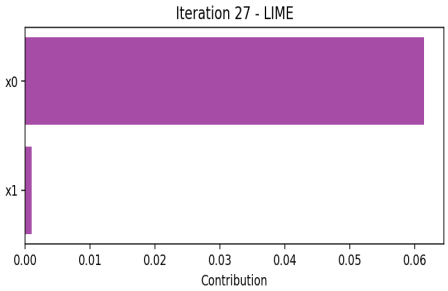
Iteration 26 - LIME Values

Feature	LIME Value
x0	0.06157
x1	0.00130

Iteration 27 - SHAP



Iteration 27 - LIME



Iteration 27 - SHAP Values

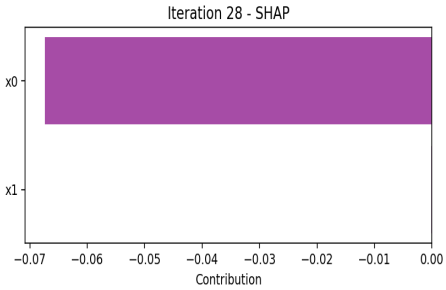
Feature	SHAP Value
x0	-0.06106
x1	-0.00033

Iteration 27 - LIME Values

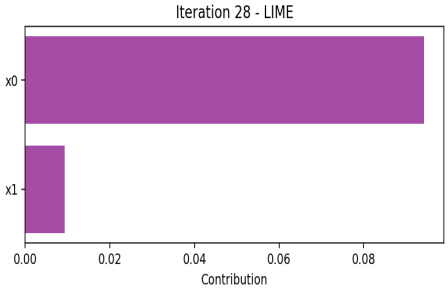
Feature	LIME Value
x0	0.06142

x1	0.00096

Iteration 28 - SHAP



Iteration 28 - LIME



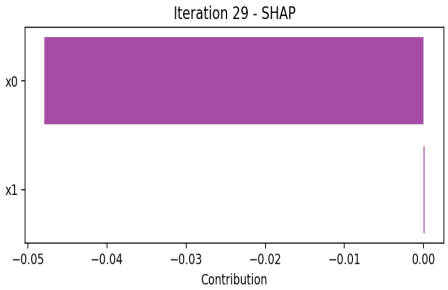
Iteration 28 - SHAP Values

Feature	SHAP Value
x0	-0.06742
x1	-0.00009

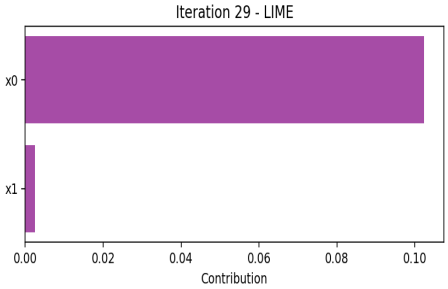
Iteration 28 - LIME Values

Feature	LIME Value
x0	0.09421
x1	0.00926

Iteration 29 - SHAP



Iteration 29 - LIME



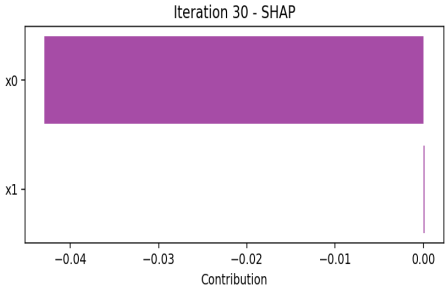
Iteration 29 - SHAP Values

Feature	SHAP Value
x0	-0.04794
x1	0.00016

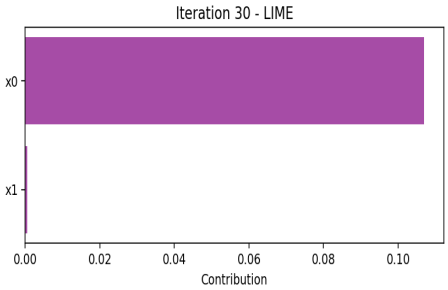
Iteration 29 - LIME Values

Feature	LIME Value
x0	0.10231
x1	0.00254

Iteration 30 - SHAP



Iteration 30 - LIME



Iteration 30 - SHAP Values

Feature	SHAP Value
x0	-0.04296
x1	0.00016

Iteration 30 - LIME Values

Feature	LIME Value
x0	0.10691
x1	0.00057